King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 261

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int \frac{3x^2+x+27}{x^2+9}\,dx =$

(a) $3x + \ln\sqrt{x^2+9} + C$

(b) $3x + \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + C$

(c) $\left(x^3 + \frac{1}{2}x^2 + 27x\right)\ln(x^2+9) + C$

(d) $3x + \tan^{-1}\left(\frac{x}{3}\right) + C$

(e) $3x + \ln(x^2+9) + C$

2. The area of the region bounded by the graphs of $f(x) = \sin x$, $g(x) = \cos x$, $x = 0$, and $x = \pi$ is equal to

(a) 2

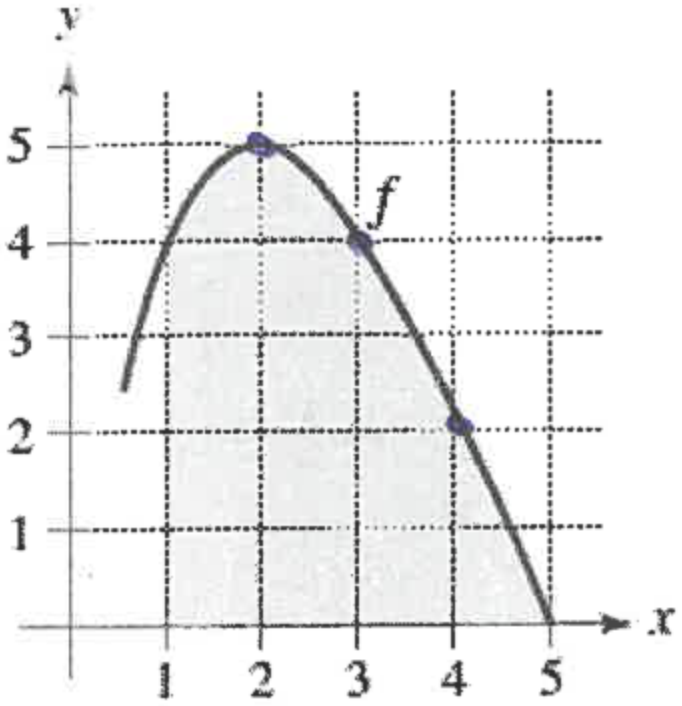
(b) $2\sqrt{2} - 1$

(c) $2\sqrt{2} + 2$

(d) $2\sqrt{2}$

(e) $2\sqrt{2} - 2$

3. The graph of $y = f(x)$ is shown below. Using rectangles of width 1, the upper sum for the region bounded by the graph of f and the x-axis on the interval $[1, 5]$ is equal to (Note: $f(4) = 2$)



(a) 10

(b) 16

(c) 17

(d) 15

(e) 11

4. If $F(x) = \int_{2\sqrt{x}}^{x} \sin(t^2)\, dt$, then $F'(4) =$

(a) $\frac{1}{2}\sin(16)$

(b) $\cos(16)$

(c) 0

(d) $\sin(16)$

(e) $\frac{1}{2}\cos(16)$

5. The area of the region bounded by the graphs of $f(x) = \dfrac{1}{1+x^2}$ and $g(x) = \dfrac{1}{2}x^2$ is equal to

(a) $\dfrac{\pi}{4} - \dfrac{2}{3}$

(b) $\dfrac{\pi}{2} - \dfrac{1}{3}$

(c) $\dfrac{\pi}{2} - \dfrac{1}{6}$

(d) $\dfrac{\pi}{4} - \dfrac{1}{6}$

(e) $\dfrac{\pi}{4} - \dfrac{1}{3}$

6. Using the Midpoint Rule with $n = 3$, the approximate area of the region bounded by the graph of $f(x) = \ln x$ and the x-axis over the interval $[1, 7]$ is equal to

(a) $\ln(105)$

(b) $2\ln(15)$

(c) $\ln(48)$

(d) $2\ln(105)$

(e) $2\ln(48)$

7. The value of c guaranteed by the Mean Value Theorem for integrals for the function $f(x) = 2 - e^x$ over the interval $[0, 3]$ is

(a) $\ln(e^3 - 4)$

(b) $\ln\left(\dfrac{e^3 - 1}{3}\right)$

(c) $\ln(e^3 - 5)$

(d) $\dfrac{7 - e^3}{3}$

(e) $3 - \ln(3)$

8. $\int 3x^2\sqrt{3x-1}\,dx =$

(a) $\dfrac{2}{7}(3x-1)^{7/2} + \dfrac{4}{5}(3x-1)^{5/2} + \dfrac{2}{3}(3x-1)^{3/2} + C$

(b) $\dfrac{1}{9}(3x-1)^{5/2} + \dfrac{2}{9}(3x-1)^{3/2} + \dfrac{1}{9}(3x-1)^{1/2} + C$

(c) $\dfrac{1}{9}(3x-1)^{7/2} + \dfrac{2}{9}(3x-1)^{5/2} + \dfrac{1}{9}(3x-1)^{3/2} + C$

(d) $\dfrac{2}{63}(3x-1)^{7/2} + \dfrac{4}{45}(3x-1)^{5/2} + \dfrac{2}{27}(3x-1)^{3/2} + C$

(e) $\dfrac{2}{63}(3x-1)^{7/2} + \dfrac{4}{5}(3x-1)^{5/2} + \dfrac{2}{3}(3x-1)^{3/2} + C$

9. $\int_2^{2-\ln 4} \sinh^2\left(1-\frac{x}{2}\right)\cosh\left(1-\frac{x}{2}\right)\,dx =$

(a) $-\frac{9}{32}$

(b) $-\frac{9}{64}$

(c) $-\frac{27}{32}$

(d) $-\frac{3}{8}$

(e) $-\frac{125}{96}$

10. $\int_{\pi/12}^{\pi/4} \csc(2x)\cot(2x)\,dx =$

(a) -1

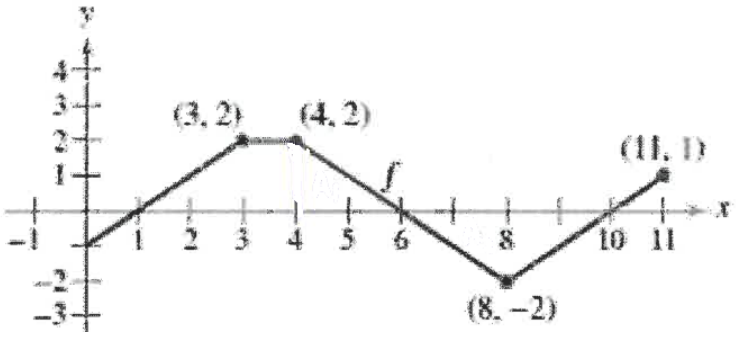
(b) $\frac{1}{2}-\frac{1}{\sqrt{3}}$

(c) $\frac{1}{2}$

(d) $-\frac{1}{2}+\frac{1}{\sqrt{3}}$

(e) $-\frac{1}{2}$

11. If the graph of $y=f(x)$ consists of line segments, as shown below, then $\int_4^{10} f(x)\,dx =$



(a) -6

(b) -2

(c) 6

(d) 12

(e) -10

12. $\int_0^{\pi/2} \tan\left(\frac{x}{3}\right) dx =$

(a) $\frac{1}{3}\ln\left(\frac{2}{\sqrt{3}}\right)$

(b) $\ln\left(\frac{\sqrt{3}}{2}\right)$

(c) $\frac{1}{3}\ln(2)$

(d) $3\ln\left(\frac{2}{\sqrt{3}}\right)$

(e) $3\ln(2)$

13. $\int_{5/2}^{15/4} \frac{1}{\sqrt{5x-x^2}}\, dx =$

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{6}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{3}$

(e) $\frac{2\pi}{3}$

14. $\int_0^{\pi/4} \frac{1}{\sin^2 x + 4\cos^2 x}\, dx =$

(a) $2\tan^{-1}(2)$

(b) $2\tan^{-1}\left(\frac{1}{2}\right)$

(c) $\frac{1}{2}\tan^{-1}\left(\frac{1}{2}\right)$

(d) $\tan^{-1}(2)$

(e) $\tan^{-1}\left(\frac{1}{2}\right)$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$= 3 + \frac{x}{x^2+9}$, so $3x + \frac{1}{2}\ln(x^2+9) + C = 3x + \ln\sqrt{x^2+9} + C$.
2	**(d)**	Split at $\pi/4$: $(\sqrt{2}-1) + (\sqrt{2}+1) = 2\sqrt{2}$.
3	**(b)**	Maxima $5, 5, 4, 2$ on the four unit intervals: sum $= 16$.
4	**(a)**	$F'(x) = \sin x^2 - \frac{\sin 4x}{\sqrt{x}}$, so $F'(4) = \frac{1}{2}\sin 16$.
5	**(b)**	Meet at $x = \pm 1$: $2\int_0^1 (\frac{1}{1+x^2} - \frac{x^2}{2})dx = \frac{\pi}{2} - \frac{1}{3}$.
6	**(e)**	$\Delta x = 2$, midpoints $2, 4, 6$: $2(\ln 2 + \ln 4 + \ln 6) = 2\ln 48$.
7	**(b)**	$2 - e^c = \frac{7-e^3}{3}$, so $c = \ln\frac{e^3-1}{3}$.
8	**(d)**	$u = 3x - 1$: $\frac{1}{9}\int (u^{5/2} + 2u^{3/2} + u^{1/2})du = \frac{2}{63}u^{7/2} + \frac{4}{45}u^{5/2} + \frac{2}{27}u^{3/2}$.
9	**(a)**	$u = \sinh(1 - \frac{x}{2})$: $-2\int_0^{3/4} u^2 du = -\frac{2}{3}\cdot\frac{27}{64} = -\frac{9}{32}$.
10	**(c)**	$u = 2x$: $-\frac{1}{2}[\csc u]_{\pi/6}^{\pi/2} = -\frac{1}{2}(1-2) = \frac{1}{2}$.
11	**(b)**	Triangle above $= 2$, triangle below $= 4$: net $2 - 4 = -2$.
12	**(d)**	$u = x/3$: $3[\ln\sec u]_0^{\pi/6} = 3\ln\frac{2}{\sqrt{3}}$.
13	**(b)**	Complete the square: $\left[\sin^{-1}\frac{u}{5/2}\right]_0^{5/4} = \sin^{-1}\frac{1}{2} = \frac{\pi}{6}$.
14	**(c)**	$u = \tan x$: $\int_0^1 \frac{du}{u^2+4} = \frac{1}{2}\tan^{-1}\frac{1}{2}$.